By the definition of adjoint, we get

$$\text{adj } A = \begin{bmatrix} (21-16) & -(-18+8) & (24-14) \\ -(-18+8) & (24-4) & -(-32+12) \\ (24-14) & -(-32+12) & (56-36) \end{bmatrix}^T = \begin{bmatrix} 5 & 10 & 10 \\ 10 & 20 & 20 \\ 10 & 20 & 20 \end{bmatrix}.$$

So, we get

$$A(\text{adj } A) = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}\begin{bmatrix} 5 & 10 & 10 \\ 10 & 20 & 20 \\ 10 & 20 & 20 \end{bmatrix}$$

$$= \begin{bmatrix} 40-60+20 & 80-120+40 & 80-120+40 \\ -30+70-40 & -60+140-80 & -60+140-80 \\ 10-40+30 & 20-80+60 & 20-80+60 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 0I_3 = |A| I_3,$$

Similarly, we get

$$(\text{adj } A)A = \begin{bmatrix} 5 & 10 & 10 \\ 10 & 20 & 20 \\ 10 & 20 & 20 \end{bmatrix}\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 40-60+20 & -30+70-40 & 10-40+30 \\ 80-120+40 & -60+140-80 & 20-80+60 \\ 80-120+40 & -60+140-80 & 20-80+60 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 0I_3 = |A| I_3.$$

Hence, $A(\text{adj } A) = (\text{adj } A)A = |A| I_3$. ■

1.2.2 Definition of inverse matrix of a square matrix

Now, we define the inverse of a square matrix.

Definition 1.2

Let A be a square matrix of order n. If there exists a square matrix B of order n such that $AB = BA = I_n$, then the matrix B is called an **inverse** of A.

Theorem 1.2

If a square matrix has an inverse, then it is unique.

Proof

Let A be a square matrix order n such that an inverse of A exists. If possible, let there be two inverses B and C of A. Then, by definition, we have $AB = BA = I_n$ and $AC = CA = I_n$.

Using these equations, we get

$$C = CI_n = C(AB) = (CA)B = I_nB = B.$$

Hence the uniqueness follows.

Notation The inverse of a matrix A is denoted by A^{-1}. ■

Note

$AA^{-1} = A^{-1}A = I_n$.

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